

Alpha or Factor Exposure?

A Replication of Fama and French (1993) and a Factor-Neutrality Test
of Industry Momentum, 1963–2026

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Abstract

A strategy that beats the market may simply be harvesting known systematic risk premia. We study this question in three steps. First, we replicate the core evidence of [Fama and French \(1993\)](#) from public data: our rebuilt SMB and HML correlate 1.0000 and 1.0000 with the published factors, the 25 size/book-to-market intercepts correlate 0.98 (CAPM) and 0.92 (three-factor) with the paper’s Table 9a, and the GRS statistics (1.98 for CAPM, 1.43 for FF3) are close to the published values. Second, we build an industry-momentum strategy (12–1 month formation, long the top 10 and short the bottom 10 of 49 industries) using only information available at the time of trade. It earns 8.63% per year ($t = 4.46$) from July 1963 to August 2026, with a CAPM alpha of 9.40% ($t = 4.90$) and an even larger Fama–French three-factor alpha of 10.78% ($t = 5.75$). Third, we neutralise known factors. Adding the momentum factor UMD raises R^2 from 0.05 to 0.67 and shrinks alpha to 2.76% ($t = 2.46$) under FF5+UMD. That residual is insignificant both in the [Fama and French](#) sample and after it, falls below the [Harvey et al. \(2016\)](#) $t > 3$ hurdle in every one of 62 UMD-controlled tests, and is eliminated by a 25 bp one-way trading cost. A control strategy, a small-value tilt, shows the textbook pattern in-sample (CAPM alpha 6.46%, FF3 alpha 0.70%). However, an unexplained FF3 alpha re-emerges after 1991 and in four international regions. We conclude that industry momentum’s apparent alpha is overwhelmingly compensation for exposure to the momentum factor, and that the three-factor model’s success is itself sample-dependent.

Keywords: asset pricing, factor models, alpha, factor neutrality, momentum, replication, out-of-sample testing.

JEL: G11, G12, G14.

1 Introduction

Every systematic strategy eventually faces one question: *is this strategy generating alpha, or am I being compensated for taking known systematic factor risk?* The distinction matters for capital allocation, fees and risk management. A return that can be reproduced by a passive combination of cheap, well-known factor portfolios is not skill, and it carries the crash risk of those factors.

The modern answer to the question was framed by [Fama and French \(1993\)](#). They showed that a market factor plus two mimicking portfolios, SMB for size and HML for book-to-market,

*Email: sakshamgarg87@gmail.com. Code, data documentation and every number in this paper are reproduced by `python scripts/run_all.py` in the accompanying repository. Data: Kenneth R. French Data Library (CRSP vintage 202608) and FRED. Code and drafting were assisted by an AI coding assistant (Anthropic’s Claude); all empirical results come from the published pipeline. Errors are mine.

capture strong common variation in stock returns. In time-series regressions of excess returns on these factors, the intercepts of most test portfolios are close to zero. The intercept, [Jensen’s alpha](#), becomes the return *not* explained by factor exposure. In [Fama and French’s](#) words, it is “the average abnormal return needed to judge whether a manager can beat the market” (Section 7.2).

This paper follows a replicate–investigate–extend design:

1. **Replication.** We rebuild SMB and HML from the six size/BE-ME building-block portfolios and re-run the paper’s regressions on its own sample (July 1963 to December 1991). We compare our estimates to the published numbers cell by cell.
2. **Investigation.** We construct a simple, transparent industry-momentum strategy and measure its raw performance. We then regress it on progressively richer factor sets (CAPM, FF3, FF3+UMD, FF5, FF5+UMD).
3. **Extension.** We test the answer out-of-sample (after 1991 and before 1963), across a parameter grid, after trading costs, across market regimes and, for a small-value control strategy, in four international regions. We also correct for the number of tests we ran.

Contribution map. We separate replication from extension and from our own analysis:

Replication	SMB/HML construction; FF93 Tables 2, 4, 6, 9a, 9c on the original 1963–1991 window; bond factors TERM and DEF (approximated).
Extension	The same tests on 1992–2026 data; FF5 and UMD benchmarks; international small-value tests; pre-1963 tests.
Own analysis	The factor-neutrality audit of industry momentum, combining: an ex-ante (look-ahead-free) factor hedge; spanning tests in both directions; regime-conditional exposures; a cost break-even; and a multiple-testing ledger of every alpha test we ran. The individual tools are standard. Our contribution is to apply them jointly and transparently to one strategy. We do not claim a new anomaly.

Preview of results. The replication succeeds closely. Industry momentum has a large, highly significant CAPM and FF3 alpha that is *not* genuine alpha in any useful sense: about two-thirds of its variance and most of its mean return come from its loading on UMD (0.87). What remains is statistically fragile and does not survive realistic trading costs. The small-value control illustrates the opposite lesson. A factor model can explain an “alpha” perfectly in the sample it was built on and fail outside it.

2 Literature

Factor models and alpha. The CAPM of [Sharpe \(1964\)](#) and [Lintner \(1965\)](#) implies that market beta alone prices assets. [Jensen \(1968\)](#) turned the intercept of a market-model regression into a performance measure. [Black et al. \(1972\)](#) introduced the time-series regression tests that [Fama and French \(1993\)](#) later extended to multiple factors. Empirically, size ([Banz, 1981](#)) and book-to-market ([Rosenberg et al., 1985](#); [Fama and French, 1992](#)) produced average returns that market beta could not explain. [Fama and French \(1993\)](#) answered with SMB and HML. [Fama and French \(1996\)](#) showed that the three-factor model absorbs many anomalies but not momentum. [Carhart \(1997\)](#) added a momentum factor, and [Fama and French \(2015\)](#) added profitability (RMW) and investment (CMA). The joint test of zero intercepts is [Gibbons et al. \(1989\)](#). Theoretical motivation for multifactor pricing comes from [Merton \(1973\)](#) and [Ross \(1976\)](#).

Momentum. Jegadeesh and Titman (1993) documented that past winners outperform past losers over 3–12 months. Moskowitz and Grinblatt (1999) showed that a large part of stock momentum is industry momentum. Grundy and Martin (2001) showed that momentum returns carry large, time-varying factor exposures, which a hedge based on ex-ante betas reduces. Cooper et al. (2004) and Daniel and Moskowitz (2016) documented that momentum’s payoff depends on market states and that momentum crashes in rebounds after bear markets. Asness et al. (2013) found value and momentum “everywhere”.

Out-of-sample decay, data mining and costs. McLean and Pontiff (2016) found that anomaly returns are 26% lower out-of-sample and 58% lower after publication. Harvey et al. (2016) argued that, given the number of factors tested, new factors need $t > 3$. Bailey and López de Prado (2014) proposed the deflated Sharpe ratio to correct for selection among many backtests. Novy-Marx and Velikov (2016) showed that trading costs eliminate many high-turnover anomalies. Lewellen et al. (2010) cautioned that test assets with a strong factor structure (such as the 25 size/BE-ME portfolios) set a low bar for asset-pricing tests. Daniel and Titman (1997) questioned whether loadings or characteristics drive the size and value premia. That question matters for whether a factor “explains” an alpha or merely shares its characteristics.

3 Hypotheses

We state hypotheses before presenting results. The industry-momentum parameters (12-month lookback, skip one month, 10 industries per leg) were fixed *a priori* from the literature convention, not chosen from our grid.

H1 (replication). In July 1963 to December 1991, FF3 explains the 25 size/BE-ME portfolios better than the CAPM: smaller absolute intercepts, higher R^2 , a lower GRS statistic. The main exception is the small-growth corner.

H2 (factor compensation, control). A long-only small-value tilt has a positive CAPM alpha that is explained by its SMB and HML loadings.

H3 (raw strategy). Industry momentum earns a positive average return and a positive CAPM and FF3 alpha.

H4 (factor neutrality, main test). Industry momentum’s alpha is *not* robust to controlling for UMD. Formally, the FF5+UMD alpha is not significant at multiple-testing-adjusted thresholds, and not positive after realistic costs.

H5 (out-of-sample). Model fit and strategy alpha deteriorate after the FF93 sample.

H6 (regimes). Industry-momentum performance and exposures depend on market states.

4 Data

All equity data come from the Kenneth R. French Data Library (French, 2026), file vintage built from the 202608 CRSP database. The bond-yield and recession series come from FRED (Federal Reserve Bank of St. Louis, 2026). Table 1 lists the series. Monthly returns are in decimal form, and the sample used for factor regressions runs from July 1963 to August 2026 (758 months).

Table 1: Data sources. “Known at” gives the earliest date at which an observation could have been used in a trading decision.

Series	Use	Known at
Mkt–RF, SMB, HML, RF	Factors; FF3	end of month t
RMW, CMA, SMB (5-factor file)	FF5	end of month t
UMD (Mom)	Momentum factor	end of month t
6 size/BE-ME portfolios (VW)	Rebuilding SMB, HML	end of month t
25 size/BE-ME portfolios (VW)	Test assets; small-value tilt	end of month t
49 industry portfolios (VW)	Strategy universe	end of month t
International 3 factors and 25 portfolios (5 regions)	International tests, from July 1990	end of month t
FRED GS10, BAA (monthly avg. yields)	TERM, DEF proxies	end of month t
FRED USREC	Recession labels	<i>months later (ex-post)</i>

Survivorship and selection. CRSP-based French portfolios include dead firms and delisting returns, so the stock universe is free of survivorship bias. The 49 industries are a fixed classification. Two industries (Health and Software) start after 1963; they enter the strategy only once they have 12 months of history. Two biases remain beyond our control. Book equity comes from Compustat, which carries backfill bias in early years; FF93 mitigate this with a two-year listing requirement that the library preserves. The library also revises historical data as CRSP and Compustat are corrected, so numbers drift between vintages (see Section 7). The repository records the SHA-256 hash and download time of every raw file.

5 Factor construction

SMB and HML. Each June, NYSE, AMEX and NASDAQ stocks are split at the NYSE median size into Small and Big, and at the 30th and 70th NYSE percentiles of book-to-market (book equity for the fiscal year ending in year $t - 1$, over December $t - 1$ market equity) into Low, Medium and High. Six value-weighted portfolios are held from July of t to June of $t + 1$. The six-month lag between fiscal year-end and portfolio formation ensures that the accounting data were public. Then

$$SMB_t = \frac{1}{3} (S/L + S/M + S/H)_t - \frac{1}{3} (B/L + B/M + B/H)_t, \quad (1)$$

$$HML_t = \frac{1}{2} (S/H + B/H)_t - \frac{1}{2} (S/L + B/L)_t. \quad (2)$$

Averaging across the other sort makes each factor approximately neutral to the other characteristic. We rebuild both from the six portfolios (Figure 3b).

TERM and DEF (approximation). FF93 define $TERM = LTG - RF$ and $DEF = CB - LTG$ using Ibbotson long-term government and corporate bond returns, which are not public. We approximate them by pricing a constant-maturity par bond off FRED yields. Buying at par at $t - 1$ with semiannual coupon $c = y_{t-1}$ and repricing at y_t gives

$$R_t = \frac{c}{y_t} \left[1 - \left(1 + \frac{y_t}{2} \right)^{-2N} \right] + \left(1 + \frac{y_t}{2} \right)^{-2N} - 1 + \frac{c}{12}. \quad (3)$$

We use $N = 20$ years (the approximate maturity of the Ibbotson long-term bond, fixed before looking at results), the 10-year Treasury yield for LTG , and Moody’s Baa yield for CB . Because

both legs use the same maturity, DEF isolates credit-spread changes. Two shortcomings follow. FRED yields are monthly averages, which smooths returns. Baa is lower grade than Ibbotson’s corporate composite, so our DEF has a higher mean. Table 6 compares the proxies with FF93.

Momentum and five-factor series. UMD is the return on high minus low prior-return (months -12 to -2) portfolios, using size-by-momentum sorts rebalanced monthly. RMW and CMA sort on operating profitability and investment (Fama and French, 2015). FF5 regressions use the five-factor file’s own SMB, which averages size sorts across three characteristics; all other regressions use the FF3 SMB.

6 Methodology

Time-series regressions. For a strategy or portfolio with excess return r_t and factor vector f_t ,

$$r_t = \alpha + \beta' f_t + \varepsilon_t. \quad (4)$$

Taking expectations, $E[r_t] = \alpha + \beta' E[f_t]$. Average return equals alpha plus each loading times its factor premium. If the factors are tradable zero-cost portfolios and the model is correct, $\alpha = 0$, because a passive combination $\beta' f_t$ replicates the strategy’s systematic return at no cost. *Factor neutrality* means $\beta = 0$. A strategy can be neutral and still worthless ($\alpha = 0$), or exposed and still valuable ($\alpha > 0$). The neutralisation question is whether $\hat{\alpha}$ survives once the relevant factors are in f_t . Equation (4) assumes constant β (we check this with rolling and regime regressions). It also assumes the factors are measured without error and that ε_t is mean-zero and uncorrelated with f_t .

Inference. Strategy regressions use Newey–West standard errors (Newey and West, 1987) with $\lceil 4(T/100)^{2/9} \rceil$ lags (Newey and West, 1994). These are robust to heteroskedasticity and autocorrelation. Replication regressions use OLS standard errors, as FF93 did. Alphas are reported in percent per year ($12 \times$ monthly).

GRS test. For N test assets and K factors over T months,

$$F = \frac{T - N - K}{N} \frac{\hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \bar{f}' \hat{\Omega}^{-1} \bar{f}} \sim F_{N, T-N-K}, \quad (5)$$

where $\hat{\Sigma}$ is the residual covariance matrix and $\hat{\Omega}$ the factor covariance matrix. The test is exact under i.i.d. normal residuals. The denominator is one plus the factors’ squared maximum Sharpe ratio. Alphas are judged relative to how good the factors already are. A unit test confirms that our implementation equals the FF93 Table 9c formula to machine precision.

Strategy and timing. At the end of month t , each industry’s signal is its cumulative return over months $t-11$ to $t-1$. We skip month t to avoid one-month reversal. We go long the 10 highest and short the 10 lowest, equal-weighted, and earn returns in month $t+1$. The P&L multiplies weights formed at t by returns at $t+1$. Unit tests verify that rewriting all data after any date leaves every earlier signal, weight and return unchanged.

Ex-ante factor hedge. The tradable factor-neutral version subtracts $\hat{\beta}'_{t-1} f_t$, with $\hat{\beta}_{t-1}$ estimated by OLS over months $t-60$ to $t-1$ on (Mkt–RF, SMB, HML, UMD). Using full-sample betas would be look-ahead bias, since the hedge ratio would contain future information. The hedge trades zero-cost factor portfolios; we do not charge costs for the hedge legs.

Transaction costs. One-way turnover is $\sum_i |w_{i,t} - \tilde{w}_{i,t}|$, where $\tilde{w}$ are last month’s weights drifted by realised returns. Net returns subtract $c \times \text{turnover}$, with c from 0 to 100 bp (base case 25 bp). Turnover inside the value-weighted industry portfolios is not charged, so costs are understated.

Out-of-sample design. The in-sample (IS) period is the FF93 window, July 1963 to December 1991 (342 months). The out-of-sample (OOS) period is January 1992 to August 2026. Neither FF93 nor the momentum literature that motivated our parameter choice could have seen it. We also test the strategy on July 1927 to June 1963, a period FF93 did not use. In the parameter-grid exercise, the specification is chosen by IS Sharpe ratio alone and judged only OOS.

Multiple testing. We log every alpha test run on an industry-momentum variant: 90 in total. We apply Holm’s step-down correction (Holm, 1979), report the Harvey et al. (2016) $t > 3$ hurdle, and compute the deflated Sharpe ratio (Bailey and López de Prado, 2014) for the grid.

7 Replication of Fama and French (1993)

Factors. The rebuilt SMB and HML correlate 1.0000 and 1.0000 with the published series over 1926–2026, with RMSE of 0.30 and 0.29 basis points per month (Table 2). Our construction formula is therefore the library’s formula. Over the FF93 window (Table 3), the market premium is 0.41% per month ($t = 1.65$; FF93: 0.43, $t = 1.76$), SMB 0.26% ($t = 1.67$; FF93: 0.27, 1.73) and HML 0.38% ($t = 2.75$; FF93: 0.40, 2.91). Correlations match: $\text{corr}(\text{SMB}, \text{HML}) = -0.09$ (FF93: -0.08), $\text{corr}(\text{Mkt}, \text{SMB}) = 0.37$ (0.32), $\text{corr}(\text{Mkt}, \text{HML}) = -0.36$ (-0.38). Small level differences reflect three decades of CRSP and Compustat revisions.

Table 2: Rebuilt vs. published factors (percent per month).

level_0	level_1	Corr	RMSE (bp/mo)	Mean rebuilt (%/mo)	Mean published (%/mo)	Months
SMB	1926-2026	1.0000	0.2964	0.1657	0.1658	1202.0000
SMB	FF93 sample	1.0000	0.2969	0.2590	0.2592	342.0000
HML	1926-2026	1.0000	0.2908	0.3518	0.3517	1202.0000
HML	FF93 sample	1.0000	0.2907	0.3773	0.3775	342.0000

Table 3: FF93 Table 2 replication, July 1963–December 1991: factor means, standard deviations (% per month) and t -statistics. TERM and DEF are our FRED-based proxies.

	Mean (ours)	Mean (FF93)	SD (ours)	SD (FF93)	t (ours)	t (FF93)
Mkt-RF	0.41	0.43	4.59	4.54	1.65	1.76
SMB	0.26	0.27	2.86	2.89	1.67	1.73
HML	0.38	0.40	2.54	2.54	2.75	2.91
TERM	0.04	0.06	2.92	3.02	0.27	
DEF	0.10	0.02	1.93	1.60	0.93	

25 size/BE-ME portfolios. Figure 1 reproduces the central finding. CAPM intercepts rise with book-to-market and fall with size, and adding SMB and HML collapses them toward zero. Mean absolute intercepts fall from 0.26% to 0.09% per month. Three-factor R^2 exceeds 0.9 for 22 of 25 portfolios (FF93: 21), against 2 under the CAPM (FF93: 2), as shown in Figure 2. The anomaly FF93 singled out reappears: the small-growth intercept is -0.37% per month ($t = -3.43$; FF93: -0.34 , $t = -3.16$) and the big-growth intercept 0.19% ($t = 2.86$; FF93: 0.21, 3.27). Across the

25 cells our intercepts correlate 0.98 (CAPM) and 0.92 (FF3) with the published ones (Figure 3a; Table 4). The mean absolute difference in FF3 intercepts is 0.04% per month.

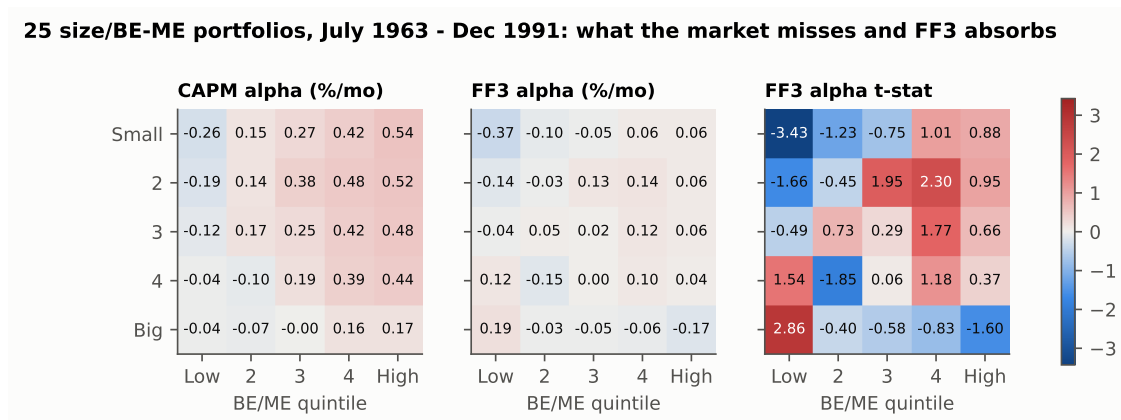


Figure 1: Intercepts of the 25 size/BE-ME portfolios, July 1963–December 1991. The CAPM leaves size and value patterns that FF3 absorbs, except in the small-growth corner.

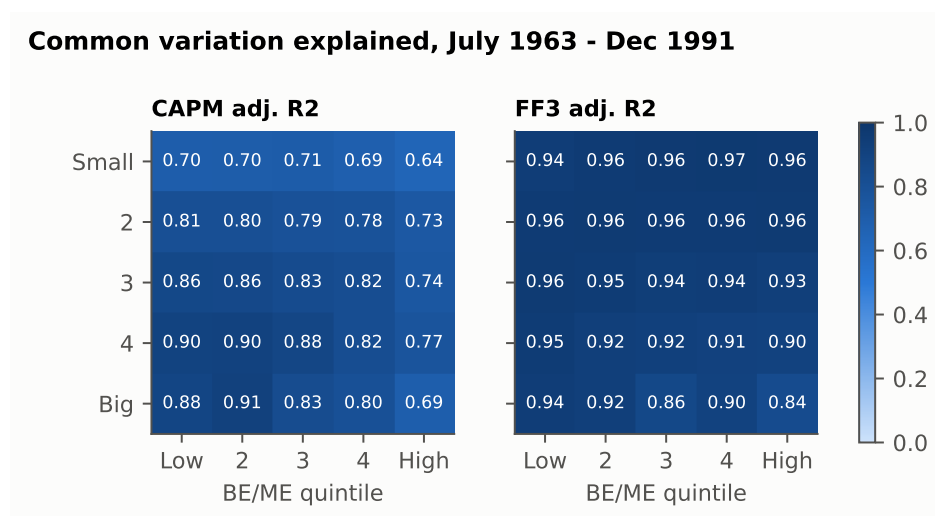
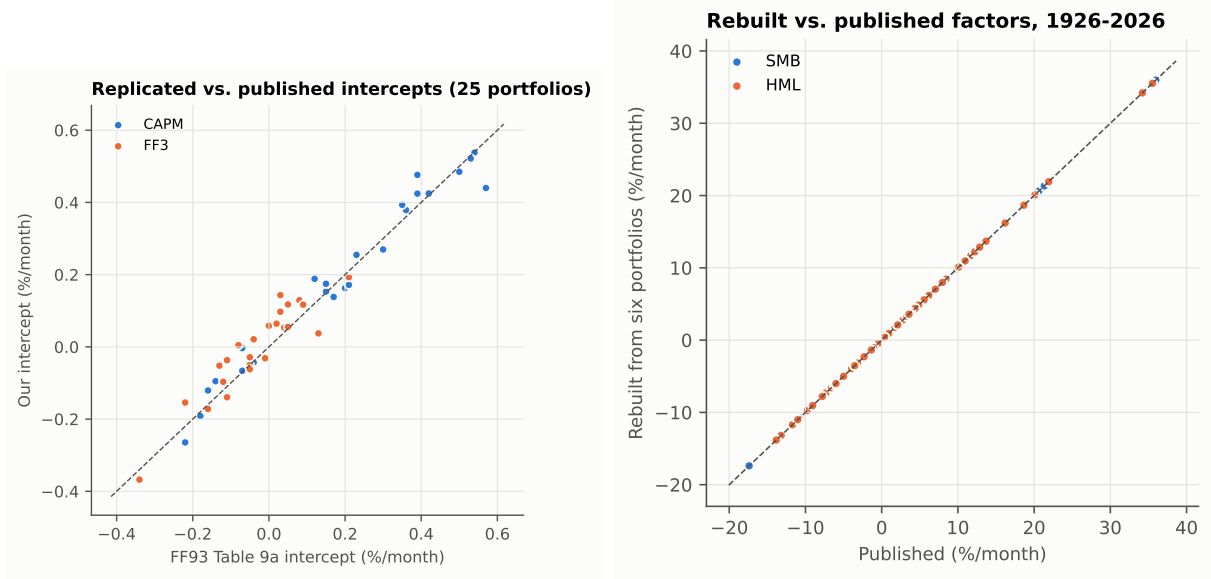


Figure 2: Adjusted R^2 , CAPM vs. FF3: SMB and HML capture strong common variation.



(a) Our intercepts vs. FF93 Table 9a.

(b) Rebuilt vs. published SMB/HML.

Figure 3: Replication accuracy.

Table 4: Cell-by-cell agreement with FF93 (25 portfolios, 1963–1991).

	Corr(ours, FF93)	Mean abs diff	Max abs diff
Mean excess	0.980	0.038	0.166
CAPM alpha	0.983	0.033	0.130
FF3 alpha	0.924	0.045	0.113
FF3 alpha t	0.874	0.658	1.791
CAPM R2	0.985	0.018	0.048
FF3 R2	0.976	0.006	0.018

GRS tests. Table 5 reports GRS statistics. In-sample, CAPM gives $F = 1.98$ ($p = 0.004$) and FF3 $F = 1.43$ ($p = 0.086$), against FF93’s 1.91 and 1.56. FF93 used 32 assets including seven bond portfolios we cannot obtain, so the comparison is approximate. The ranking is the same: TERM+DEF (2.05) worst, then CAPM, then SMB+HML, with FF3 best. Adding TERM and DEF to FF3 does not lower F (1.54), consistent with FF93’s finding that bond factors add little for stocks. Figure 21a shows the same result as average returns plotted against model-implied returns. H1 is supported.

Table 5: GRS tests of zero intercepts on the 25 size/BE-ME portfolios. FF93 column: their Table 9c (32 assets).

	FF93 F (32 assets)	F In-sample	p In-sample	a In-sample (%)	F Out-of-sample	p Out-of-sample	a Out-of-sample (%)	F Full	p Full	a Full (%)
TERM+DEF	2.090	2.055	0.003	0.547	3.935	0.000	0.612	4.278	0.000	0.551
CAPM	1.910	1.985	0.004	0.257	3.840	0.000	0.173	4.196	0.000	0.197
SMB+HML	1.750	1.538	0.051	0.505	4.016	0.000	0.761	4.284	0.000	0.613
FF3	1.560	1.432	0.086	0.094	3.713	0.000	0.116	3.643	0.000	0.087
FF3+TERM+DEF	1.660	1.536	0.051	0.095	3.799	0.000	0.117	3.581	0.000	0.086
FF3+UMD		1.457	0.076	0.103	3.446	0.000	0.109	3.219	0.000	0.080
FF5		1.111	0.328	0.083	3.343	0.000	0.120	3.228	0.000	0.088
FF5+UMD		1.209	0.228	0.092	3.116	0.000	0.113	2.935	0.000	0.080

Table 6: Bond-factor proxies, 1963–1991 (FF93: TERM mean 0.06, SD 3.02; DEF mean 0.02, SD 1.60).

	Mean (%/mo)	SD (%/mo)	corr with Mkt-RF
TERM	0.04	2.92	0.33
DEF	0.10	1.93	-0.19

8 Results: the strategy and its raw performance

Figure 4 shows the factors’ cumulative returns. Figure 5 compares the strategies with the market. From July 1963 to August 2026, industry momentum earns 8.63% per year with volatility 15.38% (Sharpe 0.56; market 0.47), $t = 4.46$, skewness -0.76 and a maximum drawdown of -53%. Its market beta is -0.11. Average one-way turnover is 87% of the long-plus-short book per month. At 25 bp the net return is 6.01% (Sharpe 0.39). The long leg beats the market and the short leg lags it (Figure 6; Table 7). The deepest drawdowns occur in 2001–2003 and in the 2009 momentum crash (a single-month loss of 34% in April 2009) (Figure 7). H3 is supported on raw returns.

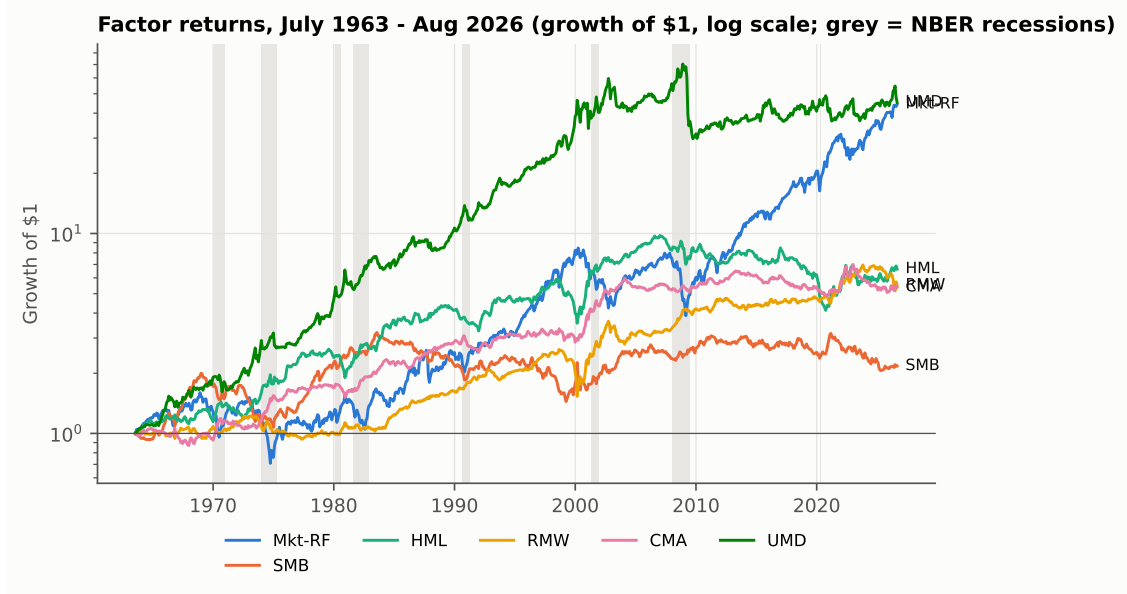


Figure 4: Cumulative factor returns, July 1963–August 2026 (log scale). Shaded: NBER recessions.

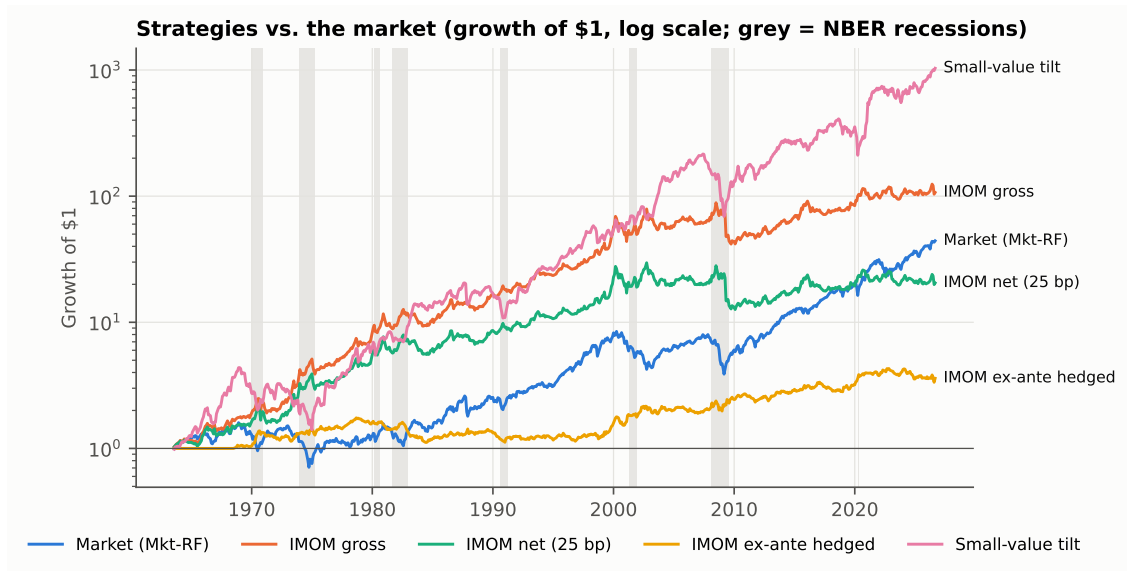


Figure 5: Strategies vs. the market. “Ex-ante hedged” removes factor exposure estimated on trailing 60-month data only.

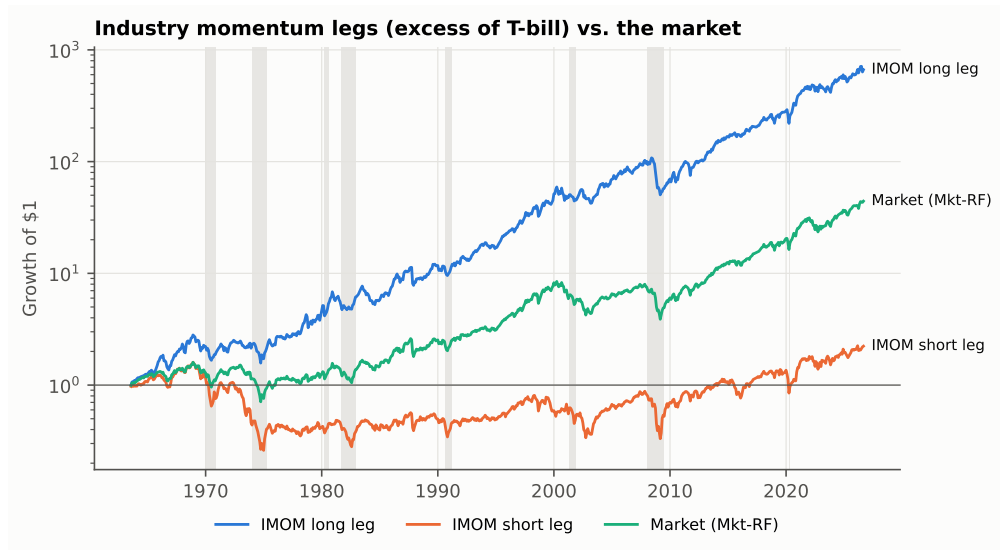


Figure 6: Industry-momentum long and short legs (excess of T-bill).

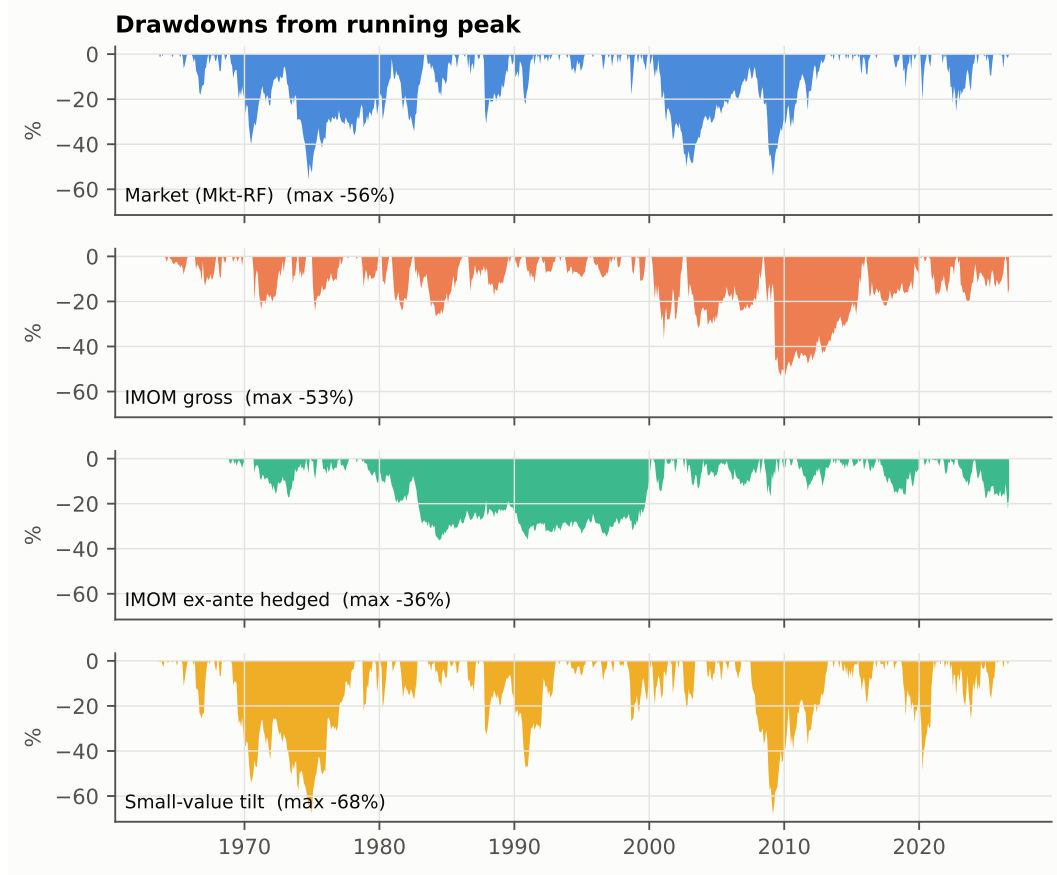


Figure 7: Drawdowns from the running peak.

Table 7: Performance summary by period (annualised). IS: 1963-07 to 1991-12; OOS: 1992-01 onward.

Period	Series	Months	Ann. mean (%)	Ann. vol (%)	Sharpe	t(mean)	Skew	Ex. kurt	Max DD (%)	Hit rate (%)
In-sample	Market (Mkt-RF)	342.00	4.93	15.91	0.31	1.65	-0.38	2.25	-55.71	54.97
In-sample	IMOM gross	342.00	11.67	14.35	0.81	4.34	-0.33	1.48	-26.29	64.04
In-sample	IMOM net (25 bp)	342.00	9.16	14.38	0.64	3.40	-0.31	1.46	-30.13	62.28
In-sample	IMOM long leg	342.00	11.18	19.59	0.57	3.05	-0.54	2.52	-43.99	57.89
In-sample	IMOM short leg	342.00	-0.49	20.82	-0.02	-0.13	0.05	2.43	-83.75	47.37
In-sample	IMOM ex-ante hedged	282.00	1.26	8.40	0.15	0.73	0.24	0.96	-36.24	52.48
In-sample	Small-value tilt	342.00	11.88	21.79	0.55	2.91	-0.00	4.07	-67.88	57.60
Out-of-sample	Market (Mkt-RF)	416.00	9.11	15.05	0.61	3.56	-0.59	1.17	-54.16	63.94
Out-of-sample	IMOM gross	416.00	6.13	16.16	0.38	2.23	-0.98	7.36	-53.00	55.29
Out-of-sample	IMOM net (25 bp)	416.00	3.41	16.18	0.21	1.24	-0.96	7.30	-57.42	52.88
Out-of-sample	IMOM long leg	416.00	12.77	17.40	0.73	4.32	-0.56	1.38	-53.34	63.94
Out-of-sample	IMOM short leg	416.00	6.64	20.56	0.32	1.90	-0.01	4.50	-62.37	57.69
Out-of-sample	IMOM ex-ante hedged	416.00	3.58	10.02	0.36	2.10	0.41	1.50	-22.21	53.61
Out-of-sample	Small-value tilt	416.00	14.60	21.58	0.68	3.98	0.23	5.42	-68.02	61.30
Full	Market (Mkt-RF)	758.00	7.22	15.45	0.47	3.72	-0.49	1.72	-55.71	59.89
Full	IMOM gross	758.00	8.63	15.38	0.56	4.46	-0.76	5.55	-53.00	59.23
Full	IMOM net (25 bp)	758.00	6.01	15.40	0.39	3.10	-0.74	5.49	-57.42	57.12
Full	IMOM long leg	758.00	12.05	18.41	0.65	5.20	-0.56	2.10	-53.34	61.21
Full	IMOM short leg	758.00	3.42	20.69	0.17	1.31	0.02	3.52	-83.75	53.03
Full	IMOM ex-ante hedged	698.00	2.64	9.40	0.28	2.14	0.38	1.50	-36.24	53.15
Full	Small-value tilt	758.00	13.37	21.66	0.62	4.91	0.12	4.81	-68.02	59.63

9 Factor attribution

The neutralisation ladder. Table 8 is the paper’s central result. The CAPM alpha is 9.40% per year ($t = 4.90$). FF3 alpha is *larger*, 10.78% ($t = 5.75$), because the strategy loads negatively on HML (-0.30): past-winner industries tend to be growth industries, and HML had a positive premium. Under FF3 alone the strategy looks like outstanding alpha with R^2 of 0.05. Adding UMD changes the picture. The UMD loading is 0.87 ($t \approx 30$), R^2 jumps to 0.67, and alpha falls to 2.31% ($t = 2.09$) under FF3+UMD and 2.76% ($t = 2.46$) under FF5+UMD (Figures 8–9).

Table 8: Industry momentum (gross), July 1963–August 2026. Coefficients with Newey–West t -statistics in parentheses; alpha in % per year.

	alpha (%/yr)	Mkt-RF	SMB	HML	RMW	CMA	UMD	adj. R^2
CAPM	9.40 (4.90)	-0.11 (-1.62)						0.01
FF3	10.78 (5.75)	-0.13 (-2.35)	-0.07 (-0.63)	-0.30 (-2.78)				0.05
FF3+UMD	2.31 (2.09)	0.03 (1.33)	-0.05 (-1.13)	-0.01 (-0.22)			0.86 (29.59)	0.67
FF5	10.42 (4.92)	-0.11 (-1.92)	-0.08 (-0.71)	-0.39 (-3.24)	-0.04 (-0.24)	0.24 (1.34)		0.05
FF5+UMD	2.76 (2.46)	0.03 (1.08)	-0.08 (-1.58)	0.01 (0.21)	-0.13 (-2.25)	-0.01 (-0.11)	0.87 (30.81)	0.67

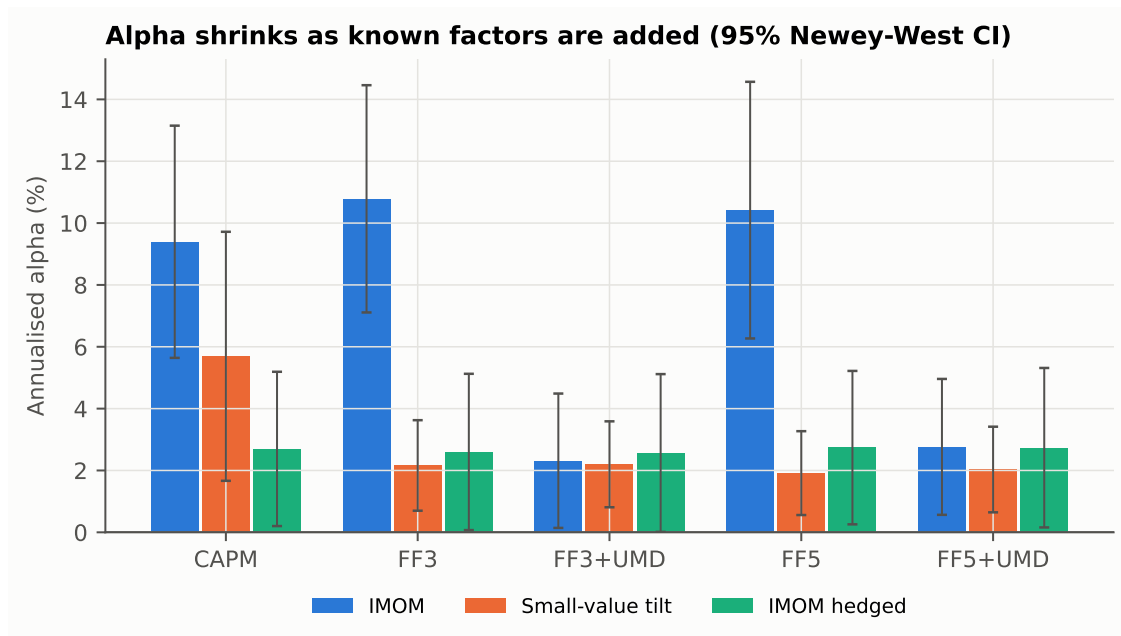


Figure 8: Alpha shrinks as known factors are added. Industry momentum loses about three-quarters of its alpha once UMD enters. The small-value tilt loses most of its CAPM alpha once SMB and HML enter.

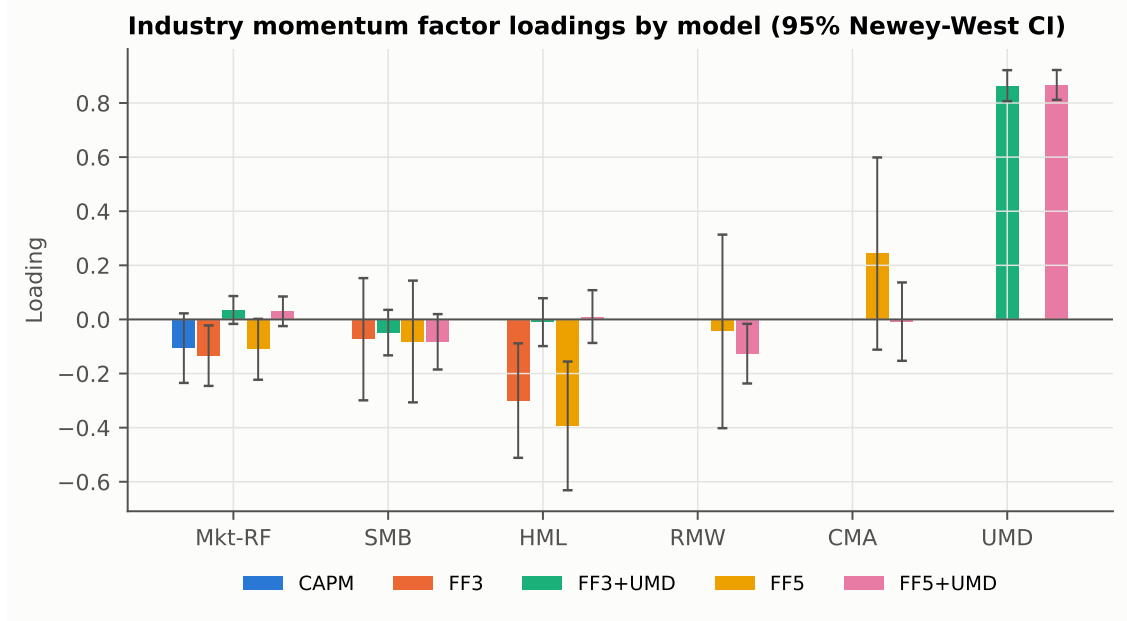


Figure 9: Industry-momentum loadings by model (95% Newey–West confidence intervals).

Risk decomposition and return attribution. Decomposing variance as $\text{Var}(r) = \beta' \Sigma_f \beta + \sigma_\varepsilon^2$, with Euler shares $\beta_k (\Sigma_f \beta)_k / \text{Var}(r)$, UMD accounts for 67% of industry-momentum variance and the residual for 33% (Figure 10). For the mean, UMD exposure contributes 6.20 percentage points per year of the 8.63% average return (Figure 11a). Figure 12 splits the cumulative return into a factor-explained part and an alpha-plus-residual part; the latter is small and noisy.

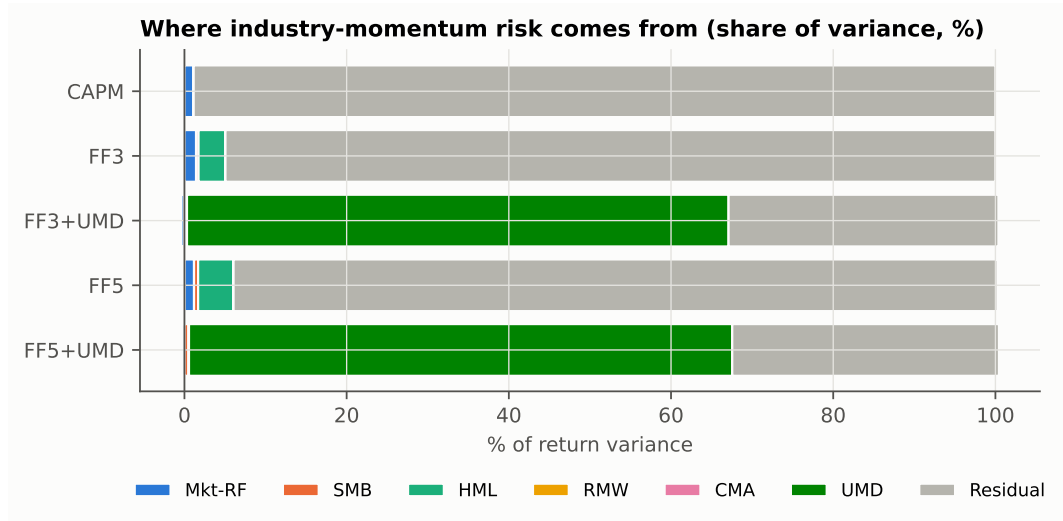


Figure 10: Share of industry-momentum return variance by source.

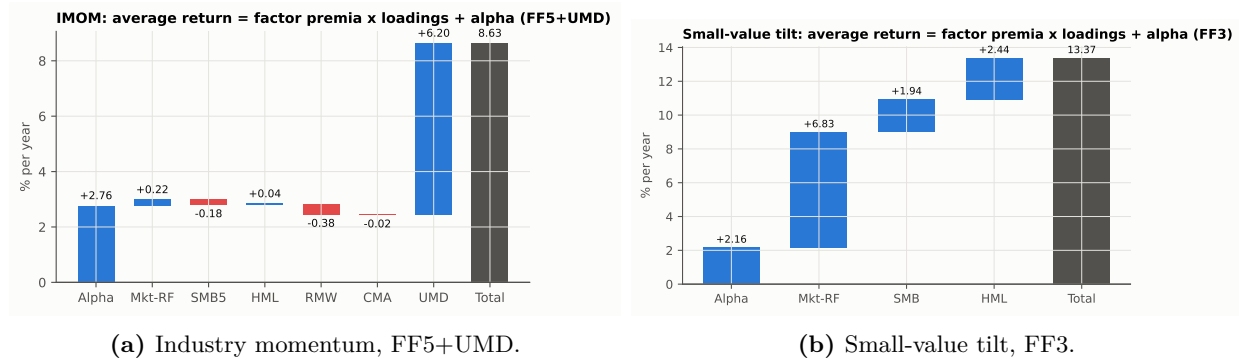


Figure 11: Average return = $\sum_k \beta_k \bar{f}_k + \alpha$ (% per year).

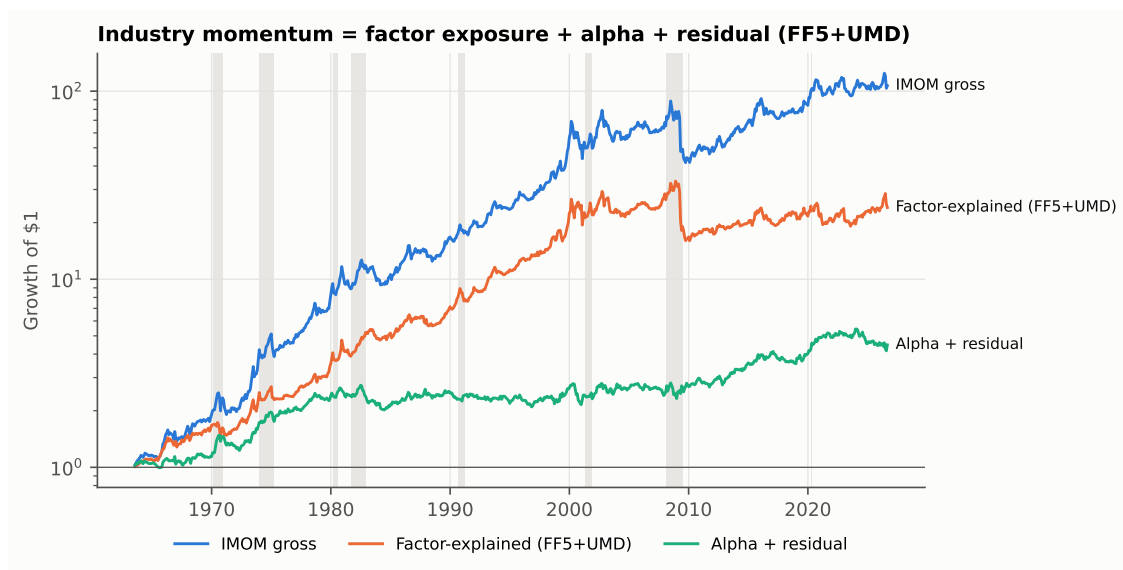


Figure 12: Industry momentum split into its FF5+UMD factor-explained component and alpha + residual.

Rolling exposures. Sixty-month rolling regressions (Figures 13–14) show that the UMD loading is persistently high, while market and value loadings swing. The rolling FF3+UMD alpha band includes zero for most of the sample. The rolling CAPM alpha is largest in the 1970s and remains positive afterwards, so the CAPM would have flagged “alpha” throughout.

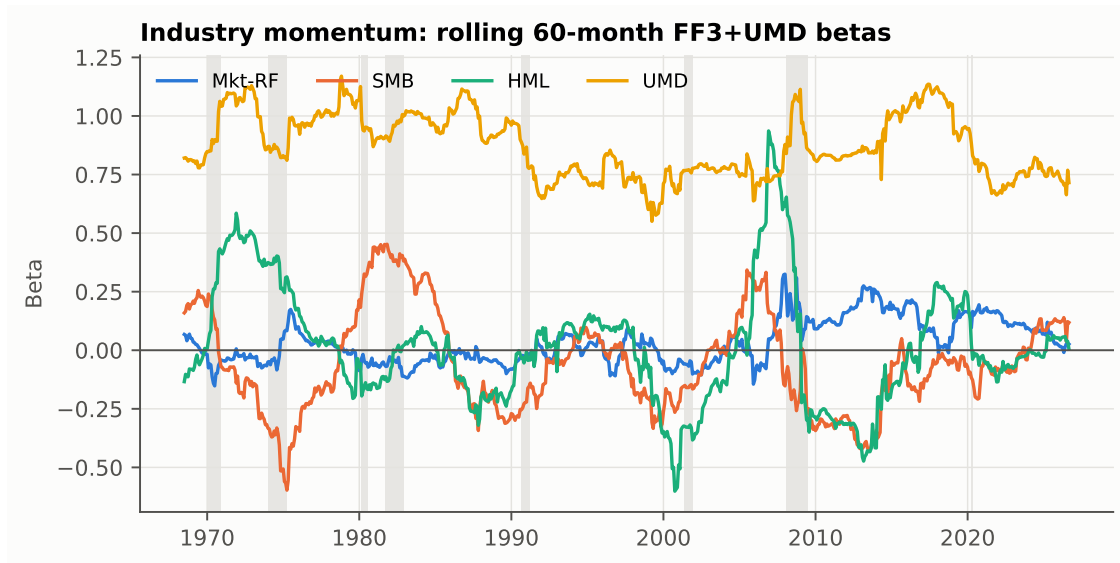


Figure 13: Rolling 60-month FF3+UMD betas of industry momentum.

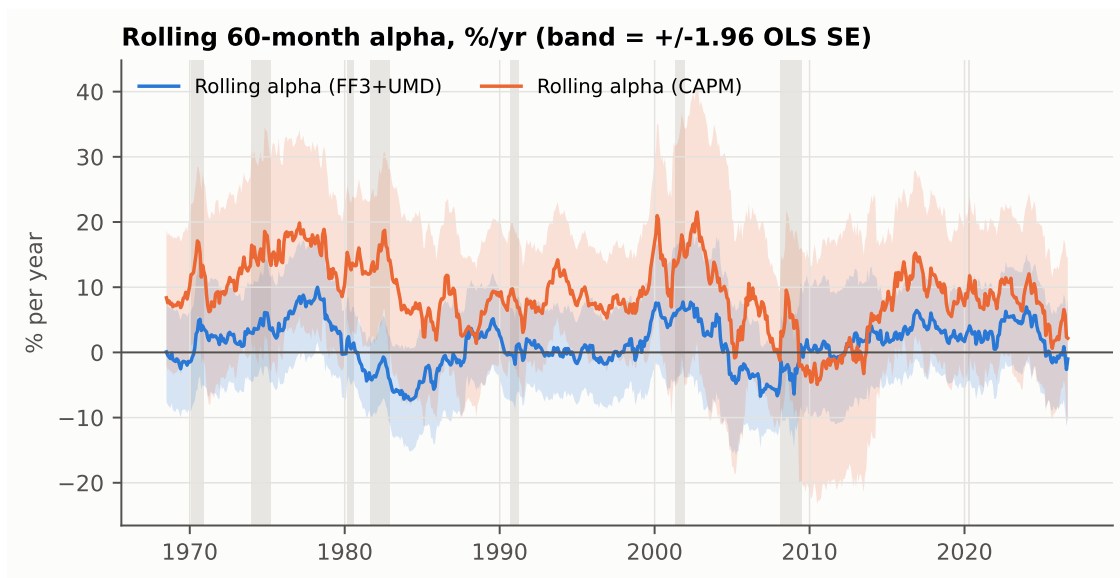
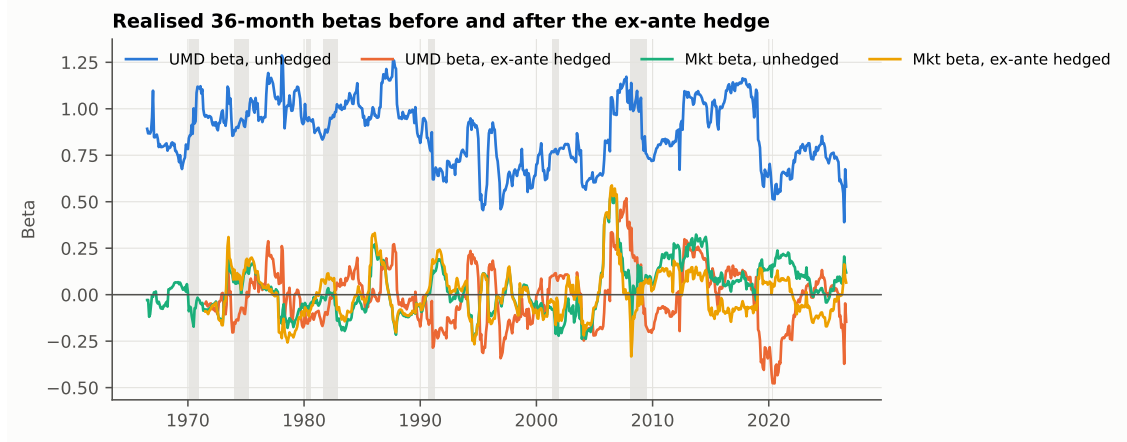


Figure 14: Rolling 60-month alpha (% per year) with ± 1.96 standard-error bands.

Ex-ante hedging. The tradable factor-neutral version earns 2.64% per year with volatility 9.40% (Sharpe 0.28). Its FF5+UMD alpha is 2.74% ($t = 2.08$; Table 9), and the hedge leaves near-zero realised loadings (Figure 15). Neutralising the exposures removes the return they earned. What is left is the same small, marginal alpha.

Table 9: Ex-ante hedged industry momentum. Newey–West t -statistics in parentheses.

	alpha (%/yr)	Mkt-RF	SMB	HML	RMW	CMA	UMD	adj. R^2
CAPM	2.70 (2.12)	-0.01 (-0.26)						-0.00
FF3	2.60 (2.02)	-0.01 (-0.28)	0.02 (0.48)	0.02 (0.44)				-0.00
FF3+UMD	2.56 (1.97)	-0.01 (-0.26)	0.02 (0.48)	0.02 (0.44)			0.00 (0.11)	-0.00
FF5	2.74 (2.16)	-0.00 (-0.13)	-0.01 (-0.19)	-0.01 (-0.11)	-0.09 (-1.30)	0.08 (0.95)		0.00
FF5+UMD	2.74 (2.08)	-0.00 (-0.13)	-0.01 (-0.19)	-0.01 (-0.10)	-0.09 (-1.31)	0.08 (1.00)	0.00 (0.01)	0.00

**Figure 15:** Realised 36-month betas before and after the ex-ante hedge.

Spanning in both directions. A regression of industry momentum on UMD cannot tell which is more primitive. Moskowitz and Grinblatt (1999) argued that industry momentum drives stock momentum. Table 10 runs the regression both ways. UMD’s alpha against FF3+industry momentum is 1.70% per year ($t = 1.54$) over the full sample, about as fragile as industry momentum’s alpha against UMD. The two strategies are largely the same risk, and neither robustly spans the other. Our evidence identifies the shared exposure, not its direction.

Table 10: Spanning tests in both directions (Newey–West t).

Period	Regression	alpha (%/yr)	t	slope on other	adj R2
In-sample	IMOM on FF3+UMD	1.43	0.84	0.99	0.68
In-sample	UMD on FF3+IMOM	2.93	2.21	0.68	0.69
Out-of-sample	IMOM on FF3+UMD	1.87	1.31	0.81	0.67
Out-of-sample	UMD on FF3+IMOM	1.61	1.09	0.79	0.68
Full	IMOM on FF3+UMD	2.31	2.09	0.86	0.67
Full	UMD on FF3+IMOM	1.70	1.54	0.75	0.68

The control: a small-value tilt. The smallest-size, highest-BE/ME portfolio earns 13.37% per year in excess of bills, with CAPM beta 1.06 and CAPM alpha 5.69% ($t = 2.77$). Its FF3 loadings are 1.10 on SMB and 0.69 on HML (Table 11; Figure 16). Over the full sample, FF3 cuts alpha to 2.16% ($t = 2.90$). The full-sample number hides a sharp split that Section 11 documents.

Table 11: Small-value tilt (S1B5 excess return), July 1963–August 2026. Newey–West t in parentheses.

	alpha (%/yr)	Mkt-RF	SMB	HML	RMW	CMA	UMD	adj. R^2
CAPM	5.69 (2.77)	1.06 (23.92)						0.57
FF3	2.16 (2.90)	0.95 (52.65)	1.10 (22.78)	0.69 (23.52)				0.91
FF3+UMD	2.20 (3.10)	0.94 (54.92)	1.10 (22.70)	0.69 (21.63)			-0.00 (-0.15)	0.91
FF5	1.91 (2.77)	0.96 (43.81)	1.08 (27.16)	0.41 (6.99)	-0.02 (-0.35)	0.24 (1.89)		0.90
FF5+UMD	2.03 (2.87)	0.96 (42.45)	1.08 (27.49)	0.41 (7.28)	-0.01 (-0.33)	0.24 (1.99)	-0.01 (-0.50)	0.90

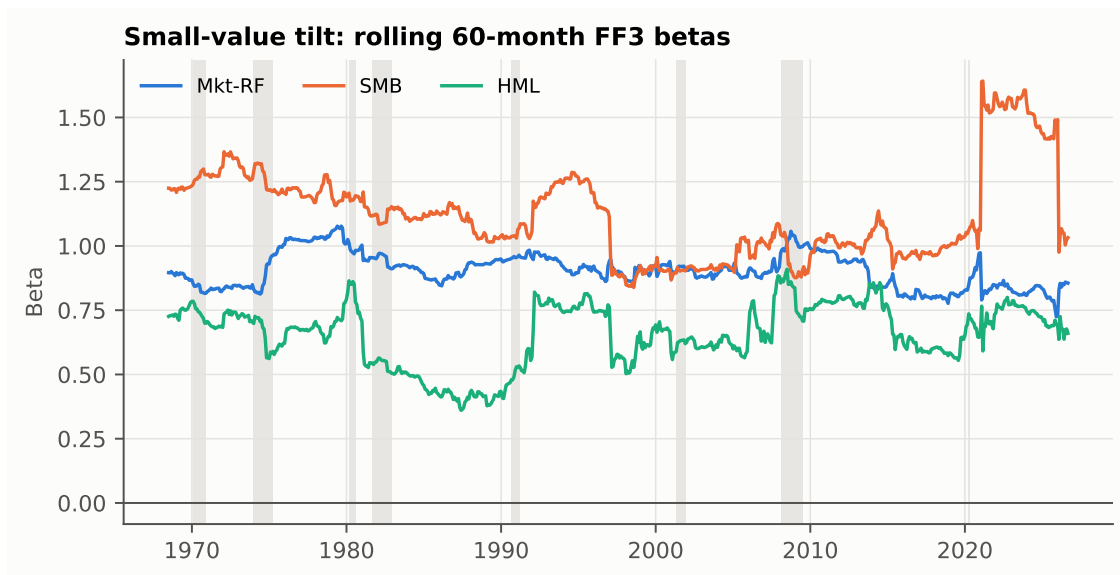


Figure 16: Rolling 60-month FF3 betas of the small-value tilt.

10 Robustness

Subperiods and the pre-1963 sample. Table 12 splits the sample. The CAPM and FF3 alphas are large in every period. The UMD-controlled alpha is insignificant in every subperiod, including 1927–1963, where industry momentum earns 5.67% per year but its FF3+UMD alpha is 0.94% ($t = 0.52$). Since 2000 the FF5+UMD alpha is 1.92% ($t = 1.16$). Excluding Januaries changes nothing.

Table 12: Subperiods (alpha in % per year; Newey–West t).

	Mean (%/yr)	Sharpe	N	CAPM alpha	CAPM t	FF3 alpha	FF3 t	FF3+UMD alpha	FF3+UMD t	FF5+UMD alpha	FF5+UMD t
1927-07..1963-06	5.67	0.33	431.00	8.58	3.76	9.47	4.02	0.94	0.52		
1963-07..1979-12	13.96	0.94	198.00	14.36	4.34	15.71	4.60	3.05	1.33	2.80	1.20
1980-01..1999-12	10.52	0.81	240.00	9.17	2.93	10.99	3.54	0.77	0.39	1.59	0.75
2000-01..	3.91	0.23	320.00	5.73	1.93	6.59	2.32	1.70	1.02	1.92	1.16
Excl. Januaries (1963-07..)	9.30	0.62	695.00	9.91	4.98	10.96	5.71	0.73	0.63	1.26	1.08

Parameter grid. Across 16 variants (lookback 3/6/9/12 months, skip 0/1, 5 or 10 industries per leg; Figure 17, Table 13), every Sharpe ratio is positive both IS and OOS. The IS and OOS rankings are correlated (Spearman $\rho = 0.72$). The IS-best specification, $L = 12$, skip = 0, $n = 10$, earns an OOS Sharpe of 0.40 (grid median 0.32) and an OOS FF5+UMD alpha of 2.93% ($t = 1.96$). Our a-priori specification ranks 2 of 16 in-sample. The deflated Sharpe ratio of the IS-best variant is 0.999, against a skill-less benchmark Sharpe of 0.27 (annualised) for the best of 16 trials. The *raw premium* is therefore not a data-mining artefact. The *alpha* is the problem.

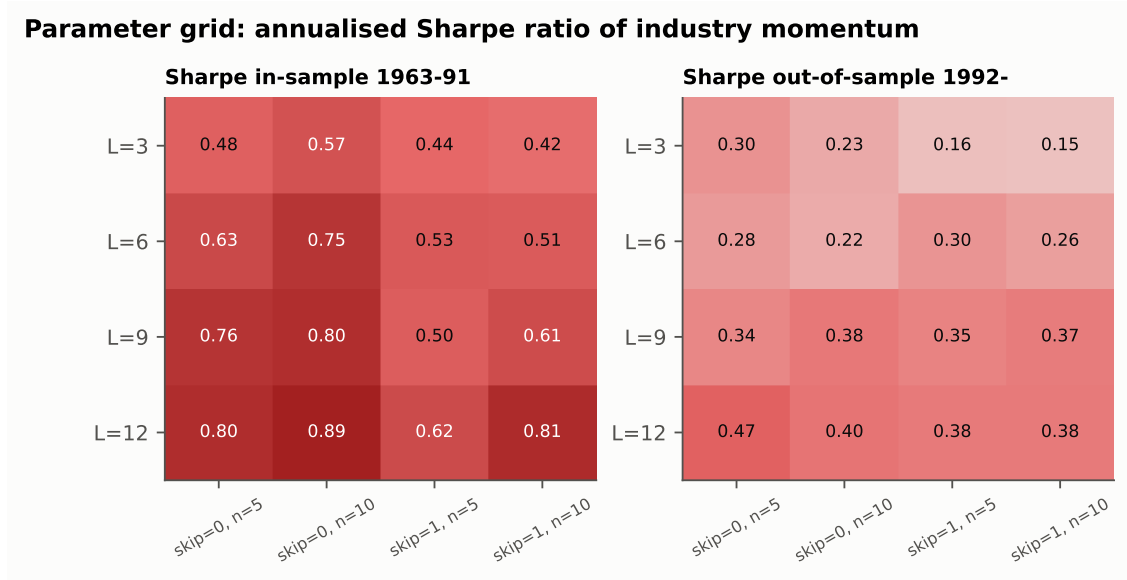
**Figure 17:** Annualised Sharpe ratio across the parameter grid, in-sample vs. out-of-sample.

Table 13: Parameter grid: Sharpe ratios and FF5+UMD alphas (% per year), IS and OOS.

lookback	skip	n	Sharpe IS	Sharpe OOS	alpha IS	t IS	alpha OOS	t OOS
3	0	5	0.48	0.30	4.32	1.25	5.78	1.69
3	0	10	0.57	0.23	3.67	1.40	3.29	1.48
3	1	5	0.44	0.16	1.69	0.51	1.12	0.33
3	1	10	0.42	0.15	-0.34	-0.14	1.02	0.44
6	0	5	0.63	0.28	4.67	1.56	2.28	0.70
6	0	10	0.75	0.22	4.01	1.88	0.69	0.29
6	1	5	0.53	0.30	0.27	0.10	1.25	0.42
6	1	10	0.51	0.26	-0.69	-0.34	0.08	0.04
9	0	5	0.76	0.34	5.13	1.69	2.49	0.87
9	0	10	0.80	0.38	2.88	1.39	2.63	1.42
9	1	5	0.50	0.35	-1.50	-0.50	2.23	0.88
9	1	10	0.61	0.37	-1.00	-0.52	1.52	0.89
12	0	5	0.80	0.47	3.51	1.23	6.19	2.23
12	0	10	0.89	0.40	3.16	1.52	2.93	1.96
12	1	5	0.62	0.38	-0.03	-0.01	2.97	1.17
12	1	10	0.81	0.38	1.82	0.98	2.26	1.60

Trading costs. Each basis point of one-way cost lowers the annual return by about 0.1 percentage points (Table 14; Figure 18). The raw premium breaks even at about 82 bp. The FF5+UMD alpha breaks even at 26 bp, and loses 5% significance at about 5 bp. Under the base 25 bp, the FF5+UMD alpha is 0.14% ($t = 0.12$). Industry-level costs for large, liquid stocks plausibly lie in the 10–30 bp range over most of the sample, and higher before the 1990s. Any genuine residual alpha is therefore within trading costs.

Table 14: Industry momentum after one-way trading costs (alpha in % per year).

cost (bp one-way)	Mean (%/yr)	Sharpe	CAPM alpha	CAPM t	FF5+UMD alpha	FF5+UMD t
0	8.63	0.56	9.40	4.90	2.76	2.46
10	7.58	0.49	8.35	4.35	1.72	1.53
25	6.01	0.39	6.78	3.51	0.14	0.13
50	3.39	0.22	4.17	2.14	-2.47	-2.16
75	0.78	0.05	1.56	0.79	-5.09	-4.40
100	-1.84	-0.12	-1.05	-0.53	-7.71	-6.56

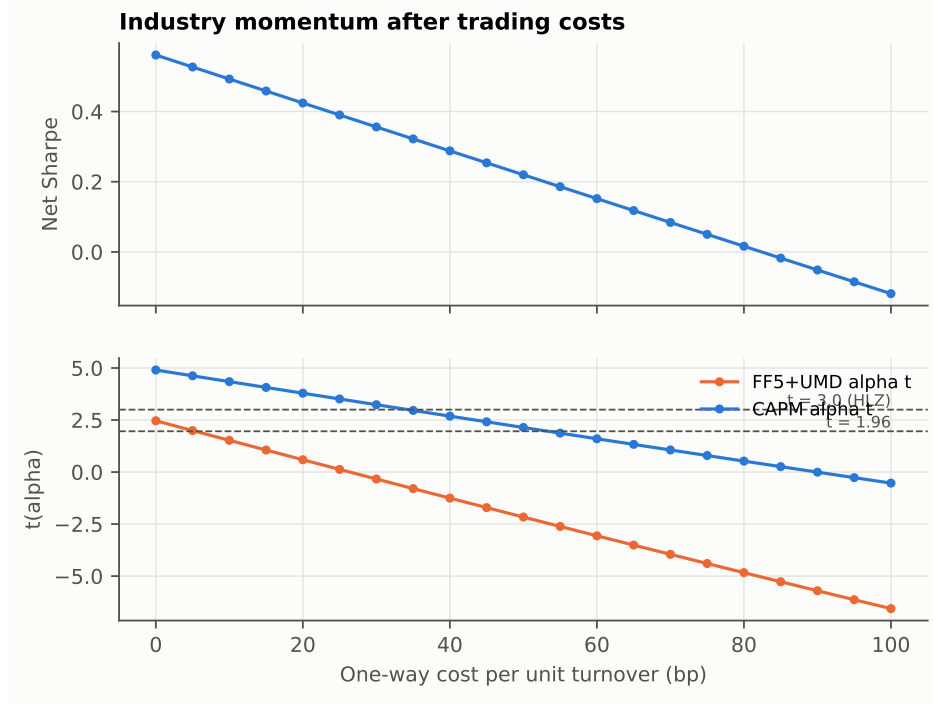


Figure 18: Net Sharpe and alpha t -statistics as costs rise. Dashed lines: $t = 1.96$ and the [Harvey et al. \(2016\)](#) $t = 3$ hurdle.

Hedge window. The ex-ante hedge with 36-, 60- or 120-month estimation windows gives FF5+UMD alphas between 1.7% and 2.7% per year, none with $t > 2.1$ (Table 15).

Table 15: Ex-ante hedge: sensitivity to the beta-estimation window.

	Mean (%/yr)	Vol (%/yr)	Sharpe	FF5+UMD alpha	t	Residual UMD beta	N
36-month window	1.62	9.71	0.17	1.66	1.33	0.02	722.00
60-month window	2.64	9.40	0.28	2.74	2.08	0.00	698.00
120-month window	1.76	9.22	0.19	1.98	1.64	0.00	638.00

Regimes. Ex-ante market states matter for the level of returns, as H6 predicts (Table 16; Figure 19). After 24-month market declines (175 months), industry momentum earns 4.75% per year, against 10.01% otherwise. In ex-ante high-volatility states it earns 6.16%, against 11.21%. The UMD loading is stable across regimes (0.82 in bear states vs. 0.85), so the regime dependence is shared with UMD itself, consistent with [Cooper et al. \(2004\)](#) and [Daniel and Moskowitz \(2016\)](#). The point estimate of FF3+UMD alpha is higher in bear states (5.92% vs. 1.68%) but imprecise. We treat it as a hypothesis for future work, not a finding. NBER recession labels are known only ex-post and are used descriptively.

Table 16: Regime-conditional performance and FF3+UMD exposures. Bear and high-volatility labels use data through $t - 1$ only.

	Months	Mean (%/yr)	Sharpe	FF3+UMD alpha	t	Mkt-RF	SMB	HML	UMD
Bear market (ex-ante): yes	175.00	4.75	0.22	5.92	1.98	-0.01	-0.16	0.08	0.82
Bear market (ex-ante): no	559.00	10.01	0.75	1.68	1.48	0.05	-0.02	-0.04	0.85
High volatility (ex-ante): yes	376.00	6.16	0.34	3.05	1.70	0.01	-0.05	-0.04	0.84
High volatility (ex-ante): no	311.00	11.21	0.91	0.92	0.66	0.07	-0.10	0.06	0.91
NBER recession (ex-post): yes	85.00	7.23	0.29	7.87	1.51	0.02	-0.25	-0.09	0.87
NBER recession (ex-post): no	673.00	8.81	0.64	1.50	1.37	0.06	-0.01	0.02	0.85

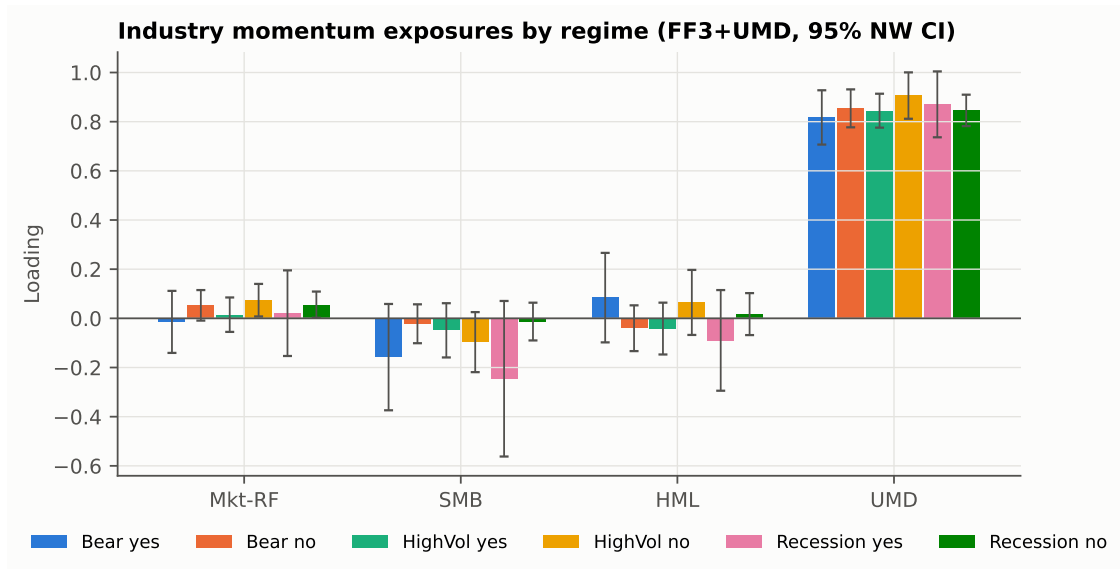


Figure 19: Industry-momentum factor exposures by regime (95% Newey–West confidence intervals).

Multiple testing. We ran 90 alpha tests on industry-momentum variants (Table in the repository, T17; Figure 20). Among the 62 tests whose benchmark includes UMD, 9 exceed $t = 1.96$, 0 exceed $t = 3$, and 0 survive Holm’s correction. Among the 28 tests with momentum-blind benchmarks, 15 survive Holm. The contrast is the paper’s answer in one figure.

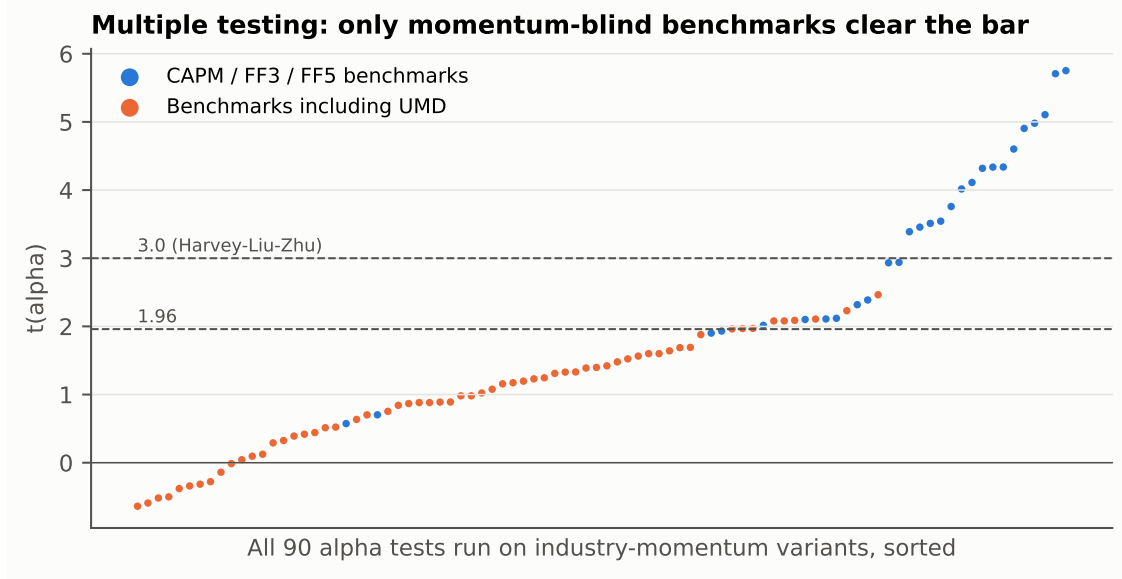


Figure 20: Every alpha test we ran on industry momentum, sorted by t -statistic. Orange: benchmark includes UMD.

11 Out-of-sample analysis

The three-factor model after 1991. The GRS test that FF3 marginally passes in-sample ($p = 0.086$) fails decisively after 1991 ($F = 3.71$, $p < 0.001$), and over the full sample ($F = 3.64$). FF5 also fails ($F = 3.23$; Table 5). Figure 21b shows why: model-implied returns cluster while realised returns disperse, with small-growth stocks far below the line.

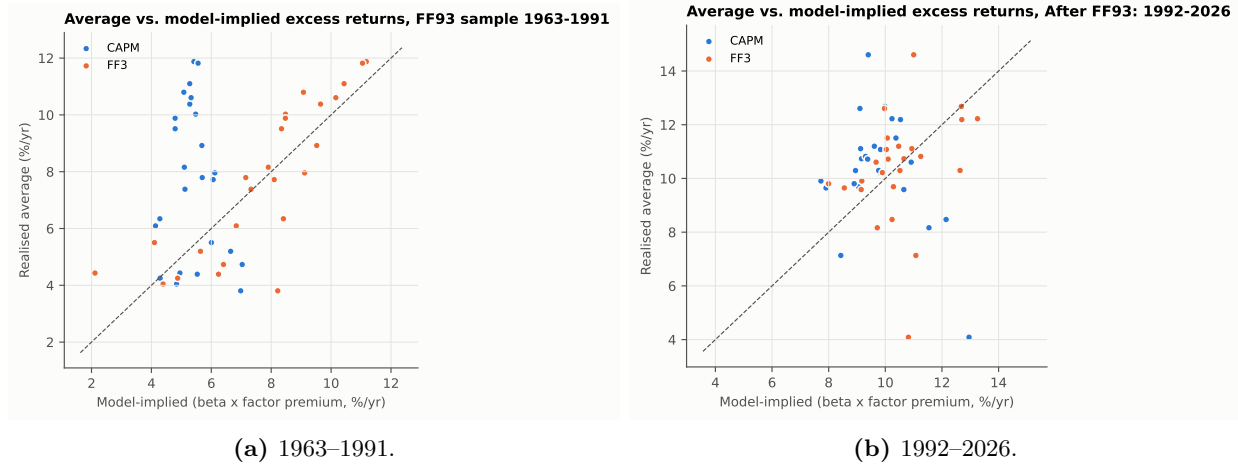


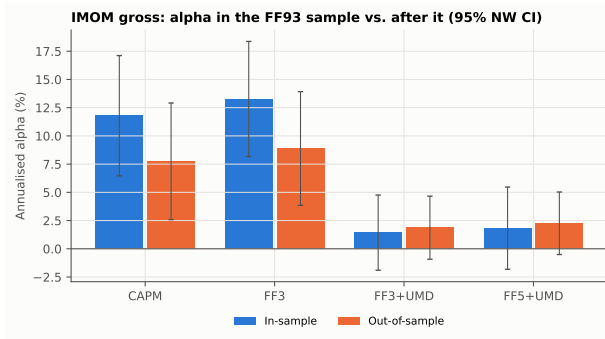
Figure 21: Average excess returns of the 25 portfolios vs. CAPM- and FF3-implied returns ($\hat{\beta}'\bar{f}$).

Strategies. Table 17 and Figure 22a compare IS and OOS alphas. Industry momentum's raw Sharpe ratio falls from 0.81 to 0.38 and its CAPM alpha from 11.79% to 7.75%. That is the post-publication decay McLean and Pontiff (2016) document, and H5 is supported. The FF5+UMD alpha is insignificant in both halves (1.82%, $t = 0.98$; 2.26%, $t = 1.60$), so the conclusion of H4 does not depend on the sample. The small-value tilt reverses the pattern (Figure 22b). In the FF93

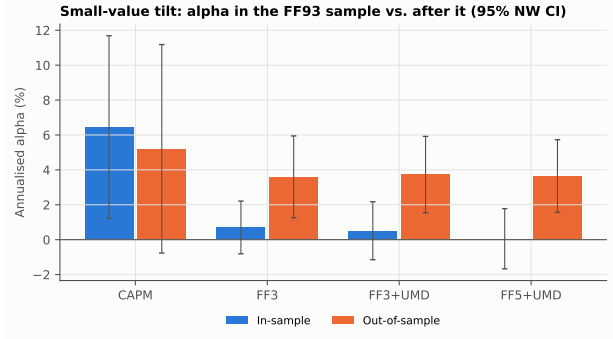
sample, FF3 explains it almost completely (alpha 0.70%, $t = 0.91$; FF5+UMD 0.05%). After 1991, an FF3 alpha of 3.60% ($t = 3.01$) appears. H2 holds in-sample only.

Table 17: Alpha (% per year) in-sample (1963–1991) and out-of-sample (1992–2026); Newey–West t in parentheses.

	CAPM	FF3	FF3+UMD	FF5+UMD
IMOM gross, IS	11.79 (4.34)	13.27 (5.11)	1.43 (0.84)	1.82 (0.98)
IMOM gross, OOS	7.75 (2.94)	8.88 (3.46)	1.87 (1.31)	2.26 (1.60)
IMOM net, IS	9.28 (3.39)	10.76 (4.11)	-1.09 (-0.64)	-0.71 (-0.38)
IMOM net, OOS	5.04 (1.90)	6.18 (2.39)	-0.85 (-0.59)	-0.45 (-0.32)
IMOM hedged, IS	1.34 (0.70)	1.08 (0.57)	1.82 (0.89)	1.95 (0.88)
IMOM hedged, OOS	3.58 (2.11)	3.56 (2.10)	3.32 (1.97)	3.57 (2.11)
Small-value tilt, IS	6.46 (2.42)	0.70 (0.91)	0.51 (0.60)	0.05 (0.06)
Small-value tilt, OOS	5.21 (1.71)	3.60 (3.01)	3.73 (3.34)	3.65 (3.43)



(a) Industry momentum.



(b) Small-value tilt.

Figure 22: In-sample vs. out-of-sample alpha by model (95% Newey–West confidence intervals).

International evidence. With regional factors from July 1990 (Table 18; Figure 23), the small-value tilt has a significant regional FF3 alpha in all 5 regions: Developed 4.50% ($t = 7.25$), North America 3.92% (5.18), Europe 2.34% (3.82), Japan 2.60% (3.84), Asia Pacific ex Japan 7.04% (5.22). FF3 t -statistics often *exceed* the CAPM ones, because the factors remove so much noise that a smaller alpha becomes more precise. This is consistent with Fama and French (2012), who find international value premiums that decrease with size and describe local factor models as only “passable” for size/BE-ME portfolios.

Table 18: Small-value tilt by region, July 1990–August 2026 (USD returns, regional factors).

	Months	Mean (%/yr)	CAPM alpha	CAPM t	GRS CAPM	GRS CAPM p	mean[a] CAPM (%/mo)	FF3 alpha	FF3 t	GRS FF3	GRS FF3 p	mean[a] FF3 (%/mo)	SMB beta	HML beta
North America	434.00	12.91	3.68	1.49	3.47	0.00	0.14	3.92	5.18	3.56	0.00	0.11	0.99	0.57
Europe	434.00	9.08	3.54	2.04	3.04	0.00	0.12	2.34	3.82	2.76	0.00	0.09	0.92	0.42
Japan	434.00	6.77	4.45	1.99	1.54	0.05	0.16	2.60	3.84	1.21	0.23	0.08	0.98	0.45
Asia Pacific ex Japan	434.00	13.93	7.00	2.52	4.08	0.00	0.22	7.04	5.22	3.57	0.00	0.20	1.19	0.53
Developed	434.00	10.91	5.34	3.07	5.54	0.00	0.14	4.50	7.25	5.22	0.00	0.11	1.00	0.46

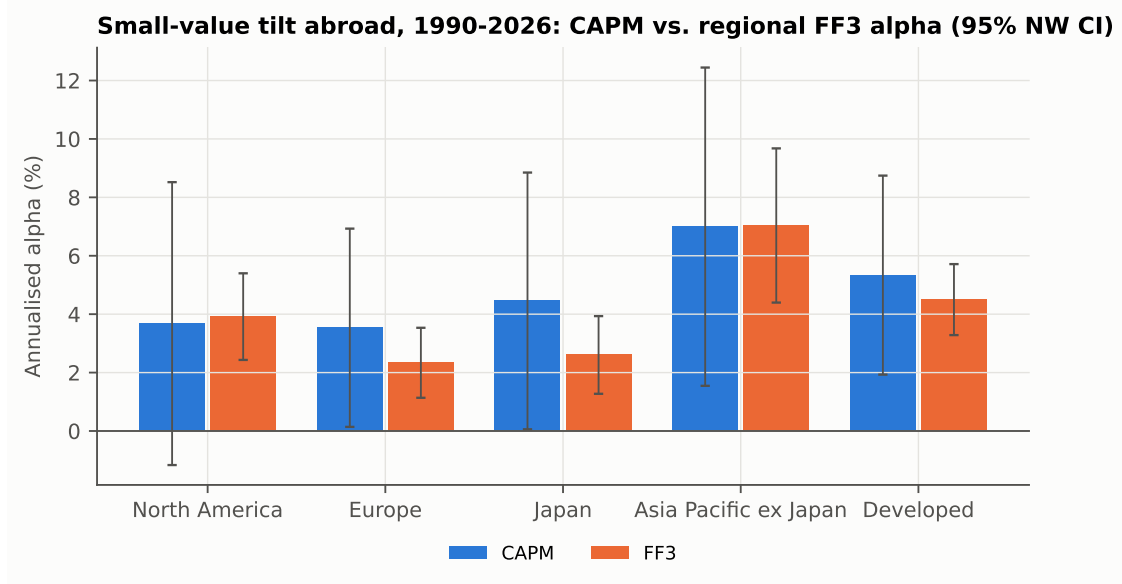


Figure 23: CAPM vs. regional FF3 alpha of the small-value tilt (95% Newey–West confidence intervals).

12 Economic interpretation

Industry momentum is a momentum-factor bet. The strategy’s apparent alpha has a simple explanation. It holds industries whose recent winners are the same stocks UMD buys, so it loads about 0.87 on UMD. UMD earned a large premium over the sample, so under any benchmark without UMD that premium looks like alpha. An investor could replicate two-thirds of the strategy’s variance and most of its mean with a passive UMD position. Paying active fees for the strategy would be paying for a factor. Whether UMD itself is a risk premium or a persistent behavioural mispricing is a separate question. Our regressions cannot answer it, because a time-series alpha of zero is consistent with both interpretations (Daniel and Titman, 1997).

Neutrality is not free. Hedging the factor exposures ex ante does exactly what it promises. The realised loadings go to roughly zero, and so does most of the return. Factor neutrality is a tool for isolating alpha, not for creating it.

A factor model is a sample-dependent benchmark. The small-value control shows the other side. FF3 explains the small-value corner almost perfectly in the sample it was designed on, then leaves a 2–7% annual alpha after 1991 and in every international region. “Explained by factors” is therefore a statement about a model in a sample. A strategy judged neutral today can show alpha against the same model tomorrow, and vice versa. Part of the post-1991 small-value alpha is probably not harvestable. The smallest NYSE size quintile contains thinly traded microcaps whose costs we do not model (Novy-Marx and Velikov, 2016).

Correlation, not causation. Every regression here measures co-movement and average returns. A UMD loading of 0.87 says industry momentum moves with stock momentum. It does not say UMD *causes* its returns, and the two-way spanning tests confirm that the data cannot order them.

13 Limitations

- **Portfolio-level data.** We trade industry portfolios, not stocks. Turnover inside industries, short-sale constraints and borrowing costs are ignored, so implementable returns are lower than reported.
- **Bond factors are proxies.** TERM and DEF come from monthly-average FRED yields with an assumed 20-year maturity. They match FF93’s qualitative role but not its exact moments, and the seven bond test portfolios of FF93 are not available. Our GRS statistics therefore use 25 assets instead of 32.
- **Data revisions.** The French library revises history. Our numbers are tied to the 202608 vintage and will drift slightly with later downloads; hashes are recorded.
- **Inference assumptions.** GRS assumes i.i.d. normal errors, which are rejected for monthly returns (fat tails, heteroskedasticity). Newey–West corrections are asymptotic. Constant betas within each regression window are an approximation.
- **Test assets.** The 25 size/BE-ME portfolios have a strong factor structure that flatters FF3 (Lewellen et al., 2010).
- **Researcher degrees of freedom.** We logged every alpha test, but choices of factor sets, windows and regime definitions are still ours. The OOS period is the only defence the data provide, and it is a single path.
- **A backtest is a hypothesis.** Nothing here establishes that any strategy will be profitable in the future.

14 Conclusion

We replicated Fama and French (1993) from public data with close agreement, then asked whether a simple industry-momentum strategy generates alpha or collects known factor premia. Against the CAPM and the three-factor model it looks like exceptional skill, with a 9–11% annual alpha and t -statistics above 4.9. Against a benchmark that includes the momentum factor, alpha shrinks by about three-quarters. The remainder fails the $t > 3$ hurdle and Holm’s multiple-testing correction, is insignificant in each half of the sample and before 1963, and disappears after 25 bp trading costs. The strategy is, to a first approximation, a levered bet on momentum. The small-value control adds a caution. A factor model can explain an alpha in its own sample and fail outside it, so “alpha versus factor exposure” is a judgement that must be re-made out of sample.

15 Future research

- Rebuild the analysis at the stock level (CRSP/Compustat) to measure true turnover, include short-sale costs and test whether *within-industry* momentum carries alpha that industry momentum does not.
- Replace TERM and DEF proxies with total-return bond indices and add the seven FF93 bond portfolios to complete the 32-asset GRS test.
- Test conditional factor models in which betas depend on ex-ante market states, and momentum-crash hedges (Daniel and Moskowitz, 2016) as candidate sources of genuine alpha.

- Extend the factor-neutrality audit to international industry momentum and to other asset classes (Asness et al., 2013).
- Use bootstrap GRS and block-bootstrap alpha confidence intervals to relax normality.

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